

SNEHA PANDIT

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EDUCATION

GLA University B.Tech Computer Science & Engineering (AI & ML) | 2023–2027

- **Relevant Coursework:** Advanced Machine Learning, Natural Language Processing, Deep Learning

RESEARCH INTERESTS

- Human-AI Interaction, Multimodal AI, Natural Language Processing, Behavioral Modeling, AI Evaluation

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, Scikit-learn, Hugging Face Transformers, Transformer Models, NLP, Deep Learning, Multimodal Learning, Vector Databases
- **Research & Evaluation:** Model Evaluation, Experimental Design, Behavioral Modeling, Human-AI Interaction, Classification Metrics
- **Programming:** Python, TypeScript, JavaScript, Java, C++
- **Systems & Infrastructure:** FastAPI, Node.js, Express, AsyncIO, Redis, Docker, GitHub Actions, Linux Environments

MACHINE LEARNING PROJECTS

[Sentia](#) | Multimodal Emotion Drift Analytics Platform | [GitHub Repository](#) | [Technical Report](#)

- Architected a parallel multimodal inference pipeline in Python using ThreadPoolExecutor, reducing P95 latency below 4 seconds.
- Fine-tuned a RoBERTa-base transformer on a normalized 7-class schema of the GoEmotions dataset (58k samples) to achieve a macro F1-score above 0.70.
- Implemented a confidence-weighted late-fusion architecture combining text embeddings with a facial expression CNN to improve multi-turn classification accuracy.
- Tracked emotional drift across multi-turn user sessions using Total Variation Distance (TVD) over 7-class probability distributions.
- Designed evaluation workflows to analyze emotional drift, behavioral consistency, and interaction safety across long-horizon human-AI conversations.

[Bhumi](#) | Smart Land Governance AI Engine | [GitHub Repository](#) | [Technical Report](#)

- Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline using LangChain to ingest, segment, and index dense, unstructured legal land records.
- Optimized similarity search routing using Hugging Face embedding models and Vector Databases to resolve domain-specific user requests with high accuracy.
- Engineered a high-throughput asynchronous backend layer using FastAPI to process concurrent document indexing streams and user queries.
- Recognized as a National Finalist at Smart India Hackathon (SIH) 2024, competing in the 36-hour Grand Finale against thousands of teams nationwide.

[Panopticon](#) | Distributed Political-Economic Data Observability | [GitHub Repository](#) | [Technical Report](#)

- Integrated a LangChain-powered processing layer into asynchronous workers to parse unstructured media feeds and perform automated entity extraction.
- Developed asynchronous data-ingestion workers via BullMQ to process over 10,000 structured events/day across distributed task queues.

OPEN-SOURCE CONTRIBUTIONS & LEADERSHIP

Project Admin / Contributor | GirlScript Summer of Code (GSSoC) | 2025–2026

- **Project Admin (2026):** Moderated code health, structured detailed contribution guidelines, and reviewed/merged contributor Pull Requests across deep learning and backend repositories.
- **Contributor (2025–2026):** Implemented modular feature extensions and refined data processing scripts across open-source AI/ML codebases

AWARDS & ACHIEVEMENTS

- **National Finalist:** Smart India Hackathon (SIH), Ministry of Education, Govt. of India | 2024
- **First Place Winner:** Smart India Hackathon Internal Nomination, GLA University | 2024
- **National Phase 2 Selection:** IBM Praambh Hack Challenge | 2025